

Valigeria (Trolley, Zaini, Borse ecc) - Luggages (Trolley, Backpack, Bags etc.)				LG
Requirements Description	Standard Method	Technical Details	Requirements	Notes
Fiber Composition [Reg.(EU) N. 1007/2011]	ISO 1833/AATCC 20/BS 4407	Method 5	±3%	1
Weight per square meter	ISO 3801	ΔE - D65 and TL84	±3%	
Color shade	ISO 105 J03 - ISO 105 A02		ΔE ≤1,0 - ≤ 4	16
Colour Fastness to Washing / Laundering	ISO 105 C06 / AATCC 61	A25 @40°-M DW	CC ≥4 ST ≥3/4	2
Colour Fastness to Perspiration	ISO 105 E04 / AATCC 15		CC ≥4 ST ≥3/4	2
Colour Fastness to Rubbing on Textiles	ISO 105 X12 / AATCC 8	W/D	Dry ≥4/5 Wet ≥3/4	15
Colour Fastness to Rubbing on Leather	ISO 11640	Dry: 150 cycles; Wet: 50 cycles	Dry ≥4 Wet ≥3	
Colour Fastness to Light	ISO 105 B02 / AATCC 16	Xenon light	≥4	
Colour Fastness to Water	ISO 105 E01	Water G3	CC ≥4 ST ≥3/4	2
Colour Fastness to Water spotting	ISO 105 E06	Wet/Dry	CC ≥4	4
Colour Fastness to Sea Water	ISO 105 E02 / AATCC 106		CC ≥4 ST ≥3/4	2
Colour Fastness to Chlorinated water	ISO 105 E03	50 ppm	CC ≥4	
Phenolic Yellowing	ISO 105 X18		CC ≥4	17
Storage migration test	JIS L0854		CC ≥4/5 ST ≥4/5	
Aging behaviour (accelerated)	Laboratory IHTM	7 days @70°C - 90%R.H.	No noticeable changes / No damages	
Dimensional Stability to Washing	EN ISO 6330	1 cycle : 5A@40°- C	±2%	5
Appearance after Laundering	Laboratory IHTM		No noticeable changes	
Movement after washing	AATCC 179	After 1 washing	±3%	14
Pilling resistance - ICI box method	ISO 12945-1	18.000 cycle	≥4	8 / 18
Pilling resistance - Martindale	ISO 12945-2	500+1.000+2.000/tex-tex	≥4	9 / 19
Seam Strength	ISO 13935-2	Fabric or sewing threads breaking	≥180N	6
Seam Slippage	ISO 13936-1	3 mm	≥160N	
Bursting Strength	ISO 13938	Mesh & Knit only	≥280kPa	
Pull resistance	EN 71-1	Decorative attachments	≥70N	12
Tensile Strength (Grab Method)	ISO 13934-2	See note 10 for details	≥180N	10
Tear Strength (Dynamometric Method)	ISO 13937	Trouser Method	see note 13	13
Tear Strength (Elmendorf Method)	ISO 13937		see note 13	13
Abrasion resistance	ISO 12947-1 & 2	12 kPa - 2 ends	≥20.000 cycles - CC ≥3/4 @5.000 cycles	19
Water permeability (Hydrostatic head)	ISO 811	Water proof fabrics only	≥ 1.800 mm	7
Water repellency (Spray test)	EN ISO 4920	after 1 washing cycle	ISO 4	20
Flex abrasion resistance on prints	JIS L1096	500 rotations	No damages	
Slide fasteners (Zips)	EN 16732 (BS 3084)	See Table note 23	see Table note 23	23
Buckles open/close strength	EN 71-1	Pull test after 500 open/close manual cycles	≥200N	
Pull strength resistance of Rings and Buckles	EN 71-1	Pull test on 5 specimens	≥300N	
Snap buttons pull opening strength	EN 71-1	Pull test on 5 specimens	18N ÷ 36N	
Touch and Close Fasteners - Longitudinal Shear Strength	ISO 13780		6.5 N/cm² after 5.000 Cycles	22
Shoulder strap and Handle strength	EN 71-1	Pull test on 5 specimens	≥300N	
Shaking and impact resistance (Jerk)	Laboratory IHTM	Handles : 1000 cycles Trolley sistem: 500 cycles Testing load accordingly with Table (see Note)	No damages (functional and visual)	(27
Pull-rod fatigue resistance	Laboratory IHTM	3000 cycles open/close	No damages (functional and visual)	(
Tumbling resistance	Laboratory IHTM	Testing load accordingly with Table (see Note) - 50 cycles	No damages (functional and visual)	(27
Millage resistance	Laboratory IHTM	16 or 32 Km @4km/h speed (see note for more details) 1 stop every 4 km for 10min	No damages (functional and visual)	(24 / 25 / 27
Drop resistance	Laboratory IHTM	Testing load accordingly with Table (see Note) Nr. of drop times : 2 each side / angle Height Drop: 100 cm	No damages (functional and visual)	26 / 27

Notes

- No tolerance for single fibre [Reg.(EU) N. 1007/2011, Art.20 comma 3] . Other standard than mentioned can be applied for fiber content determination.
- Trims & Color contrast: ST ≥4-5
- ST in the solvent
- Combined evaluation with Spray Test
- Flat dry - Knitt and Mesh : ± 4% for
- Test must be carried out on the seams of the bags
- Applicable to fabric with Water proof treatment .
- Assessment after 7.000 cycles, if ≥ 3/4 perform additional 11.000 cycles and evaluate again.
- Assessment after 500 cycles and after 1000 cycles
- Requirement not applicable for knitt fabric, Polyester/Nylon/Canvas/PU: ≥180N, Leather:≥110N, Lining: ≥150N
- After 1 washing (5A@40°-C)
- Laminated, Sewn-on, Metal branding e.g. Logo
- Test and requirement not applicable for knitt fabric and Mesh
- Option 1
- For Pigment Prints & Denim : Dry≥3/4 Wet≥3
- PP sample vs Bulk or Shipment sample. ISO 105 J03 for instrumental evaluation. ISO 105 A02 for visual evaluation
- All White/Light Colors/Fluorescent
- ICI box pilling test method applicable to knitt only. Martindale pilling test method applicable to woven fabric only .
- Evaluation of appearance at 5.000 cycles shall be descriptive.
- After 1 washing cycle
- Applicable to PU and PVC
- Shear strength before 5.000cycle (initial) : 7.5 N/cm²
- Requirements for Zipper (Slide Fastner)

Slide fastners (Zips) requirements					
Zipper size	#5	#5	#8	#10	
Performance code	C	D	D	E	EN 16732
Reciprocating			≥ 3000 cycles		
Strength of top stop	≥ 90 N	≥ 110 N	≥ 110 N	≥ 130 N	EN 16732
Lateral strength of slide fastner	≥ 250 N	≥ 370 N	≥ 370 N	≥ 470 N	EN 16732
Torque strength of slider	≥ 0,51 Nm	≥ 0,60 Nm	≥ 0,66 Nm	≥ 0,70 Nm	EN 16732
Zipper teeth pull strength	≥ 50 N	≥ 50 N	≥ 80 N	≥ 90 N	EN 16732
Strength of puller attachment	≥ 200 N	≥ 250 N	≥ 250 N	≥ 300 N	EN 16732

- Testing "distance" : Spinner cases 16 km Upright & Horizontal cases: 32 km
- Applicable to Spinner cases only (rollers)
- Standard testing conditioning 25°C - Special Cycle Testing. Conditioning Up on request only

Drop test Conditioning	
-30°C	Uncontrolled R.H.
0°C	85%
25°C	50%
40°C	85%

- Load for testing vs max nominal maximum load to be claimed on the product. For shaking and impact resistance (Jerk) test on Trolley maximum testing load is 20 kgs .

Load for testing vs size		
	Size	Load for testing (kgs)
Messengers	S	3,0
	M	4,0
	B	5,0
	S	4,0
Backpacks	M	6,0
	B	15
	S	1,0
	M	2,0
Shoulder Bags	D	5,0
	S	7
	M	15
	B	25

Legenda

- CC Color Change
ST Color Staining
Stre Stretch
RE Residual Extension
ISO International Standard Organization
BS British Standard
EN European Norm
AATCC American Association of Chemist and Colourist
IHTM In House Test Method